CET 479 Van Nguyen
Spring 2023
In class Activity – Feb 27, 2023

LAB_06 : Shell Script

1. Write a shell script to display 'this is CET479' output on the command line.

```
echo "This is CET479"     : display a message
```

$ [chmod +x cet479.sh] : Save the file with a .sh extension

$ [./cet479.sh] : Run the script of the message "this is CET479"

2. Now to the above script append 2 # signs. Check the output. Did it change?
   To append use the following
   cat >> scriptname
   #
   #

Ctrl +d to exit.

```
#!/bin/bash
   echo "This is CET479"     : display a message
   #
   #
```

```
┌──(van㉿kali)-[~]
└─$ cat>>myscript

 #!/bin/bash

 echo " This is CET479 "

#
#

┌──(van㉿kali)-[~]
└─$
```

$ [cat>>cet479.sh]

#

#

The # character at the beginning of each line indicates that the line is a comment and will be ignored by the script interpreter.

3. Create a variable NAME=yourname. Use the same script to display the output. Again, you can use the append function. Show that your script works.

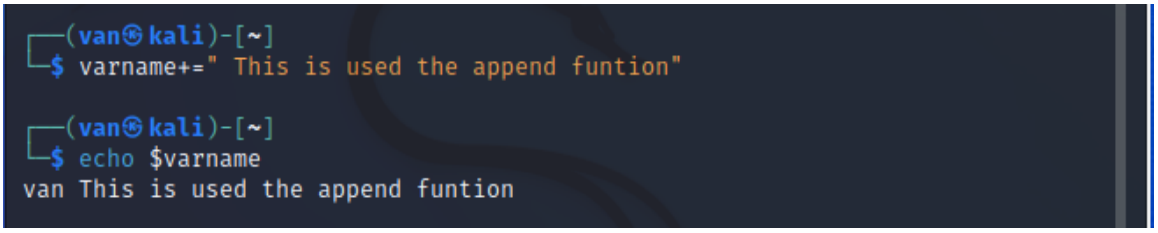
$ [varname="Van"] : To declare a variable named "varname" with the value "Van"

```
┌──(van㉿kali)-[~]
└─$ varname="van"

┌──(van㉿kali)-[~]
└─$
```

$ [**echo** **$**varname] : To display the value of a shell variable.

$ [**varname+=** "This is used the append function"] : add a string to a the end of variable "varname"



4. Use the vim editor to practice creating a script to display output of a variable (any variable of your choice) and then run your script. Show the output here.

$ [**vim myscript**] : Creating a file vim editor => open page of file

- Use key " **i** " for enter text (insert).

  #!/bin/bash

  echo " Hello everyone class CET 479 Spring 2023 "
- Use key " **esc** " exit the mode insert .
- Use " **:wq** " save and quit the page enter data.

$ [chmod +x myscript] : To be able to run the script without specifying the interpreter from the command line you'll need to make the file executable

$ [./ myscript] : run the file myscript

```
[vannguyen@fedora ~]$ vim myscript
[vannguyen@fedora ~]$ chmod +x myscript
[vannguyen@fedora ~]$ ./myscript
 Hello everyone class CET 479 spring 2023
[vannguyen@fedora ~]$
```

Note: For the following questions 5, 6 and 7, you do not have to write a script. You can test in your command line interface directly.

5. Create a temporary directory and a file within it. Type ls to see that the file exists. Now type the following:

```
[vannguyen@fedora ~]$ mkdir TEMPORARY
[vannguyen@fedora ~]$ cd TEMPORARY
[vannguyen@fedora TEMPORARY]$ touch question_5
[vannguyen@fedora TEMPORARY]$ ls
question_5
[vannguyen@fedora TEMPORARY]$
```

$filelist=$(ls) : Assigns the output of the ls command to the variable filelist

$echo $filelist : Prints the value of the variable filelist to the terminal. The $ sign before the variable name is used to reference its value.

```
[vannguyen@fedora TEMPORARY]$ filelist=$(ls)
[vannguyen@fedora TEMPORARY]$ echo $filelist
question_5
[vannguyen@fedora TEMPORARY]$
```

What do you observe?

the output is name of file stored in directory currents the variable .

6. Test the following expressions and write the outputs that you observe:

```
$ value= expr 1 + 2
$ echo $value
```

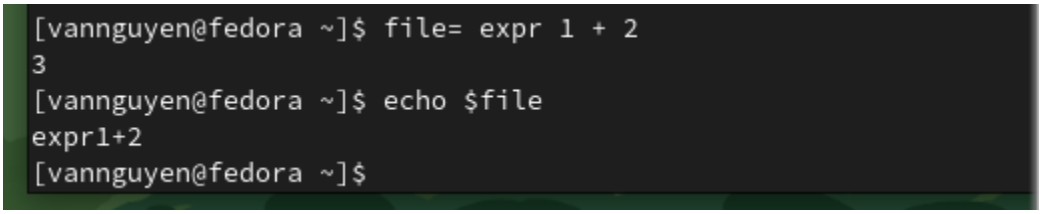

```
$ value ="expr 1 + 2"
$ echo $value
```


```
$ value = 'expr 1 + 2'
$ echo $value
```


```
$ value = ` expr 1 + 2 `
$ echo $value
```

7. Now in your terminal type the following commands:
   file=1+2
   echo $file

```
$ file= expr 1 + 2
$ echo $file
```

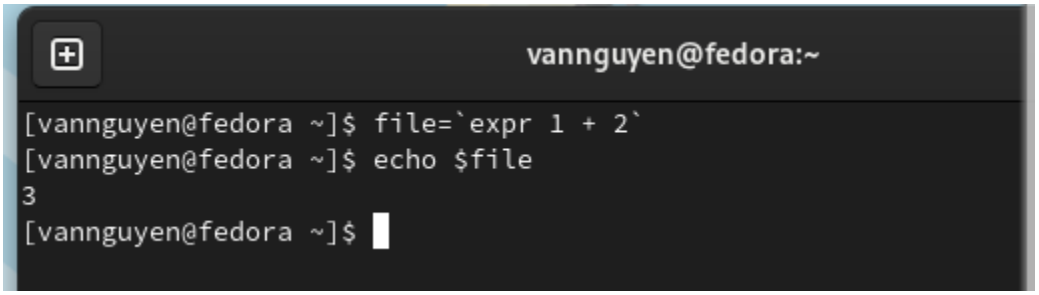


What did you observe?

Result of expression 1 + 2

Can you fix the commands to display sum of 1 & 2.

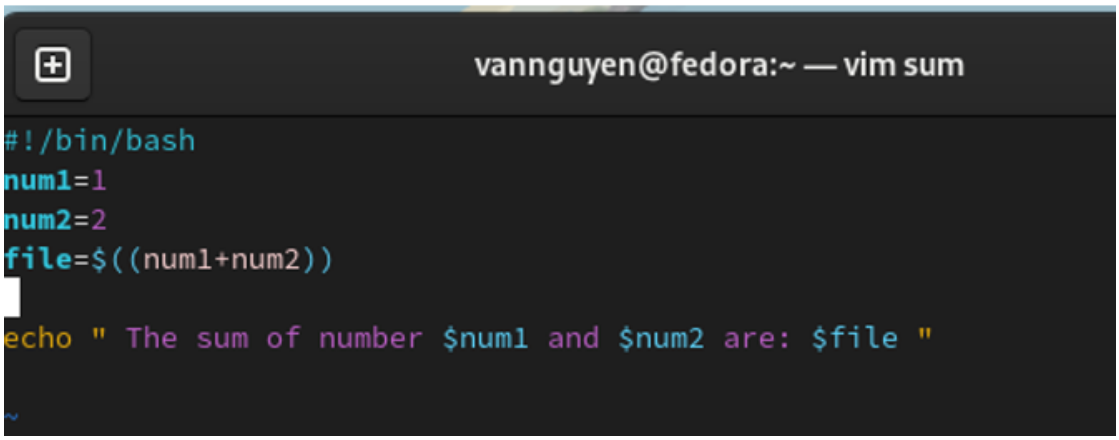
```
$ file=`expr 1 + 2`
$ echo $file
```



8. Using the vim editor, create a script to print the sum of two numbers on the screen. The output should be properly displayed - using sentences like “the numbers are … and …”. and “the sum is……”

$ [vim sum] : Creating a file vim editor => open page of file type :

```
#!/bin/bash
num1=1
num2=2
file=$((num1+num2))
echo "The sum of number $num1 and num2 are : $file"
```



$ [chmod +x sum]

$ [./ sum] : run the file sum